

Differentiating hyperbolic trig functions

Key Points

To differentiate any hyperbolic trig function use the exponential or logarithm form for inverses. Useful ones to remember are

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

An alternative for inverses is to rearrange the equation

Basic Skills

Q1. Write down the exponential form of $\sinh 2x$

Q2. Draw the graph of $y = \coth x$

Q3. State $\frac{d}{dx}(e^{kx})$

Q4. Find the gradient of $y = 5 \sinh x$ at $x = 0$

Q5. Find the gradient of $y = \tanh 3x$ at $x = 2$

Q6. Determine the stationary point and classify it for the graph $y = \cosh x$

Q7. By using the exponential form prove the derivative of hyperbolic cosine $\frac{d}{dx}(\cosh x) = \sinh x$

Competency Skills

Q8. Prove that $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$ by

a) Using the identity of $\tanh x$ **b)** Using the exponential form of $\tanh x$

Q9. Find $\frac{d}{dx}(\sinh^{-1} x)$ by

a) Using writing $x = \sinh y$ **b)** Using the logarithm form $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$

Q10. Show that $y = \sinh x \cosh x$ does not have any stationary points

Q11. Find the tangent to the function $f(x) = \operatorname{sech} x$ when $x = 1$

Q12. Show that $\frac{d}{dx}\left(\operatorname{arccosh}\left(\frac{x}{a}\right)\right) = \frac{1}{\sqrt{x^2 - a^2}}$

Q13. Show that $\frac{d}{dx}\left(\operatorname{arctanh}\left(\frac{x}{a}\right)\right) = \frac{a}{a^2 - x^2}$

Advanced Skills

Q14. Find $f'(x)$ for $f(x) = \operatorname{arcsinh} x^2$

Q15. Find the values of k such that $f(x) = e^{kx} \cosh x$ has stationary points

Q16. Find the derivative of the function $y(x) = x^2 \sinh(x^2)$

Q17. For $y = \operatorname{arcsech}\left(\frac{x}{4}\right)$

a) Show that $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 16}}$

b) Hence find the exact value in logarithm form of $\int_1^2 \frac{5}{x\sqrt{64 - 4x^2}} dx$

Extension

Q13. Explain why proving multiple argument identities is easy with hyperbolic trig than standard trig